

Co-operative Genetic Pooling and Individual Genetic Selection

A Unified Theory of Bioeconomic Evolution,
Information Capital, and Multi-Species Optimization

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Abstract

This paper develops a general theory of co-operative genetic pooling and individual genetic selection. Genetic information is modeled as a non-rival informational capital stock distributed across biological species. A common genetic information pool expands the feasible evolutionary opportunity set, while decentralized selection preserves species-level optimization. The framework integrates biology, bioengineering, economics, finance, biostatistics, machine learning, political science, biopolitics, and international relations into a single mathematical theory of evolutionary information allocation. Evolution is reinterpreted as a dynamic information economy operating over a multi-species genetic capital market.

The paper ends with “The End”

1 Introduction

The history of life may be interpreted as a process of information accumulation, transmission, modification, and selection.

Traditional evolutionary theory focuses primarily upon competition, selection, and adaptation. Modern biotechnology introduces the possibility of directed information transfer across biological boundaries.

The present paper investigates a generalized question:

What are the mathematical, economic, and political consequences of treating genetic information as a globally distributed informational capital stock?

We propose a two-level framework:

1. Co-operative genetic pooling.
2. Individual genetic selection.

The resulting theory resembles a hybrid of evolutionary biology, portfolio theory, information economics, and political economy.

2 Fundamental Structure

2.1 Species Space

Let

$$S = \{s_1, s_2, \dots, s_n\}$$

denote the set of species.

For each species,

$$G_i$$

denotes its genetic information set.
Define the global information space

$$G = \bigcup_{i=1}^n G_i.$$

Definition 1 (Desired Genetic State). *For each species s_i define*

$$D_i \subseteq G$$

as the desired genetic information set.

The genetic deficit is

$$\Delta_i = D_i \setminus G_i.$$

The quantity

$$|\Delta_i|$$

measures informational scarcity.

3 Genetic Information Geometry

Define the distance

$$d(A, B) = \frac{|A \Delta B|}{|A \cup B|}.$$

This induces a metric over informational states.
Species similarity becomes

$$d_{ij} = d(G_i, G_j).$$

Low values imply evolutionary proximity.

4 Co-operative Genetic Pooling

Suppose species contribute information:

$$C_i \subseteq G_i.$$

The common pool is

$$P = \bigcup_i C_i.$$

Accessible information becomes

$$A_i = G_i \cup P.$$

The post-pooling deficit is

$$\Delta'_i = D_i \setminus A_i.$$

Definition 2 (Pooling Efficiency).

$$E_i = 1 - \frac{|\Delta'_i|}{|\Delta_i|}.$$

5 Individual Genetic Selection

Access does not imply adoption.

Let

$$x_{ig} \in [0, 1]$$

represent selection intensity.

Define

$$X_i = (x_{ig})_{g \in G}.$$

The realized genome is

$$R_i(X_i).$$

Species maximize

$$F_i(X_i),$$

where F_i denotes fitness.

6 Portfolio Theory of Genetics

Genes are treated as assets.

Expected contribution:

$$\mu_g.$$

Variance:

$$\sigma_g^2.$$

Covariance matrix:

$$\Sigma.$$

Expected fitness return:

$$\mu_i = \sum_g x_{ig} \mu_g.$$

Risk:

$$V_i = X_i^T \Sigma X_i.$$

Optimization:

$$\max_X (\mu_i - \lambda_i V_i).$$

This is formally identical to Markowitz portfolio allocation.

7 Genetic Capital Theory

Definition 3 (Genetic Capital).

$$K_i = |G_i|.$$

Enhanced capital:

$$K_i^* = |R_i|.$$

Growth:

$$g_i = \frac{K_i^* - K_i}{K_i}.$$

7.1 Production

Let

$$Y_i = A_i K_i^\alpha L_i^\beta T_i^\gamma.$$

where

- L_i = population,
- T_i = technology,
- A_i = efficiency.

8 Bioeconomic Equilibrium

Assume genetic information possesses market value.

Let

$$p_g$$

denote the shadow price of genetic information element g .
Species wealth:

$$W_i = \sum_{g \in G_i} p_g.$$

Equilibrium satisfies

$$Q_g^D = Q_g^S.$$

for every information element.

9 Genetic Finance

Future benefit stream:

$$C_{gt}.$$

Present value:

$$V_g = \sum_{t=0}^{\infty} \frac{E(C_{gt})}{(1+r)^t}.$$

This interprets genes as informational securities.

9.1 Genetic Options

Define

$$H_g = \max(F_g - K, 0).$$

This behaves as a call option on future evolutionary success.

10 Network Theory

Construct graph

$$\mathcal{G} = (V, E).$$

Vertices:

$$V = S.$$

Weighted edges:

$$w_{ij}.$$

Information diffusion:

$$\dot{G}_i = \sum_j w_{ij} (G_j - G_i).$$

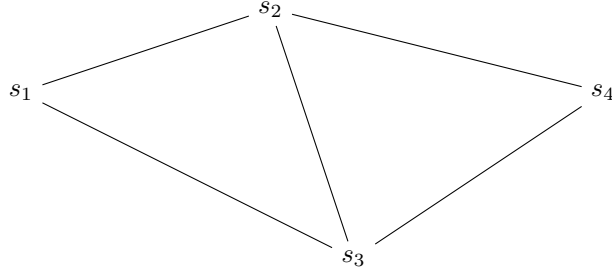


Figure 1: Genetic Information Exchange Network

11 Machine Learning Interpretation

Let

$$\theta_i$$

represent the genomic parameter vector.

Learning objective:

$$\max_{\theta_i} F_i(\theta_i).$$

Evolution resembles distributed optimization.

The update equation becomes

$$\theta_i^{t+1} = \theta_i^t + \eta \nabla F_i.$$

Pooling corresponds to federated learning.

Selection corresponds to local model optimization.

12 Biostatistical Foundations

Suppose trait value

$$Y$$

depends upon genetic factors:

$$Y = X\beta + \varepsilon.$$

Heritability:

$$h^2 = \frac{\sigma_G^2}{\sigma_P^2}.$$

Genetic pooling alters

$$\sigma_G^2$$

through expansion of accessible information.

13 Biopolitics

Control over genetic information implies control over future biological capability.

Define political genetic power:

$$P_i = \phi(K_i, T_i, L_i).$$

where

- genetic capital,
- biotechnology,
- population

jointly determine influence.

14 International Relations

States may be viewed as custodians of biological information assets.

Three strategic regimes emerge.

1. Genetic openness.
2. Genetic protectionism.
3. Genetic imperialism.

The strategic payoff matrix resembles technology competition.

Policy	Innovation	Security
Open Pooling	High	Low
Protectionism	Medium	High
Imperial Control	High	Unstable

Table 1: Strategic Bioeconomic Regimes

15 Philosophical Foundations

The theory suggests a distinction between:

1. Information availability.
2. Information realization.

Pooling expands possibility.

Selection determines actuality.

This parallels Aristotle's distinction between potentiality and actuality.

16 Fundamental Theorem

Theorem 1 (Co-operative Genetic Pooling). *Assume:*

1. *Genetic information is non-rival.*
2. *Pooling cost is finite.*
3. *Species retain selection autonomy.*

Then

$$|A_i| \geq |G_i|$$

for every species.

Furthermore,

$$\sum_i |\Delta'_i| \leq \sum_i |\Delta_i|.$$

Proof. Since

$$A_i = G_i \cup P,$$

we have

$$G_i \subseteq A_i.$$

Therefore

$$|A_i| \geq |G_i|.$$

Next,

$$D_i \setminus A_i \subseteq D_i \setminus G_i.$$

Thus

$$|\Delta'_i| \leq |\Delta_i|.$$

Summing over all species yields

$$\sum_i |\Delta'_i| \leq \sum_i |\Delta_i|.$$

□

17 Pareto Frontier

Global optimization:

$$\max \sum_i \omega_i F_i.$$

Subject to

$$R_i \subseteq D_i$$

and

$$E(R_1, \dots, R_n) \leq E_{\max}.$$

This defines a frontier between

1. species fitness,
2. ecosystem stability,
3. informational diversity,
4. political control.

18 Discussion

The theory transforms evolution from a purely competitive process into a mixed system containing both:

1. co-operative information expansion,
2. decentralized selective optimization.

Economic growth, machine learning, capital accumulation, and biological evolution become manifestations of a common information-theoretic process.

19 Conclusion

Co-operative genetic pooling enlarges the accessible informational frontier. Individual genetic selection determines realized biological states. Together they generate a universal framework for understanding bioeconomic evolution, information capital formation, biotechnology, and the political economy of life itself.

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Glossary

S	Species space.
G_i	Genetic information set.
G	Global genetic information space.
D_i	Desired genetic information set.
Δ_i	Genetic deficit.
P	Common genetic pool.
A_i	Accessible information set.
X_i	Genetic portfolio.
F_i	Fitness function.
K_i	Genetic capital.
Y_i	Bioeconomic output.
W_i	Genetic wealth.
V_g	Present value of genetic information.
w_{ij}	Exchange intensity.
P_i	Political genetic power.

The End